

## LAB EXPERIMENTS

**17-07-2026**

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### **Set 1: Monthly Sales Data**

#### **Dataset: Monthly Sales**

- 1. Create a line chart to visualize the monthly sales. Label the axes and title the chart appropriately.**

#### **Code :**

```
# Load library
```

```
library(ggplot2)
```

```
# Dataset
```

```
sales <- data.frame(
```

```
  Month = factor(c("Jan", "Feb", "Mar", "Apr", "May", "Jun",  
                  "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"),
```

```
                  levels = c("Jan", "Feb", "Mar", "Apr", "May", "Jun",  
                             "Jul", "Aug", "Sep", "Oct", "Nov", "Dec")),
```

```
  Sales = c(12000, 15000, 18000, 17000, 22000, 25000,  
            24000, 27000, 30000, 32000, 35000, 40000)
```

```
)
```

```
# Line Chart
```

```
ggplot(sales, aes(x=Month, y=Sales, group=1)) +
```

```
  geom_line(color="blue", linewidth=1.2) +
```

```
  geom_point(color="red", size=3) +
```

```
  labs(
```

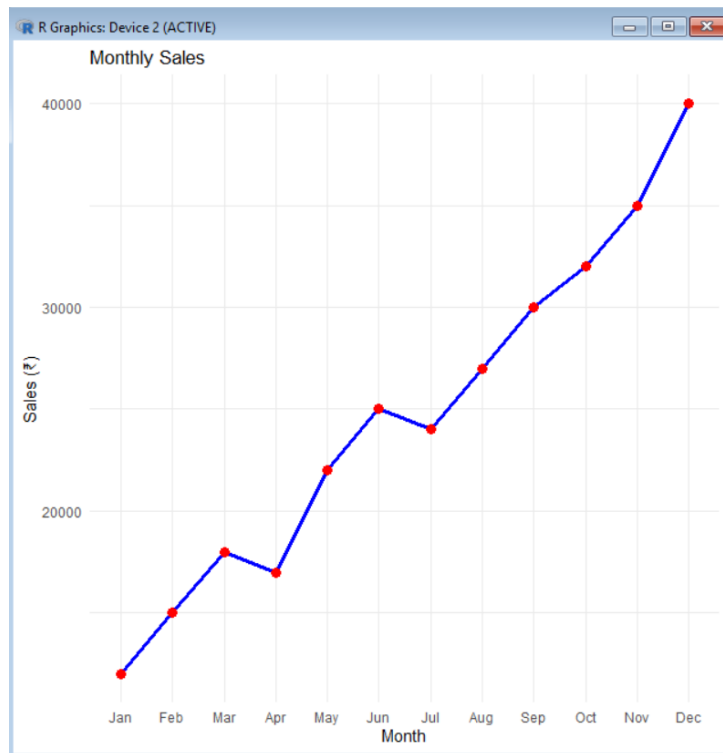
```
    title="Monthly Sales",
```

```

x="Month",
y="Sales (₹)"
) +
theme_minimal()

```

OUTPUT :



**1.B .Generate a bar chart to display the top-selling products for the year. Label the chart elements.**

**CODE :**

```

# Load library
library(ggplot2)

# Dataset
products <- data.frame(
  Product=c("Laptop","Mobile","Tablet","Headphones",
            "Smartwatch","Camera"),
  Sales=c(450,700,320,500,280,210)
)

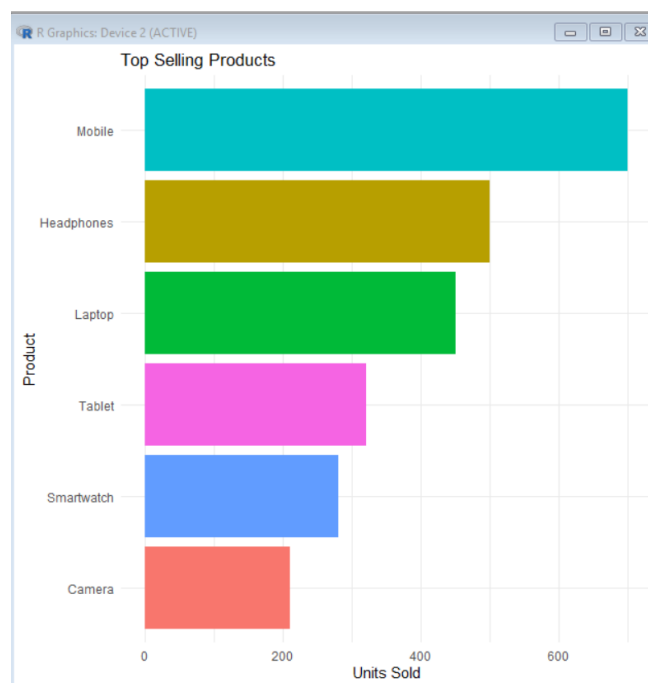
```

)

# Bar Chart

```
ggplot(products, aes(x=reorder(Product, Sales), y=Sales, fill=Product)) +  
  geom_bar(stat="identity") +  
  coord_flip() +  
  labs(  
    title="Top Selling Products",  
    x="Product",  
    y="Units Sold"  
  ) +  
  theme_minimal() +  
  theme(legend.position="none")
```

OUTPUT :



**1.C. Develop a scatter plot to explore the relationship between advertising budget and monthly sales.**

**Explain the insights drawn from the scatter plot.**

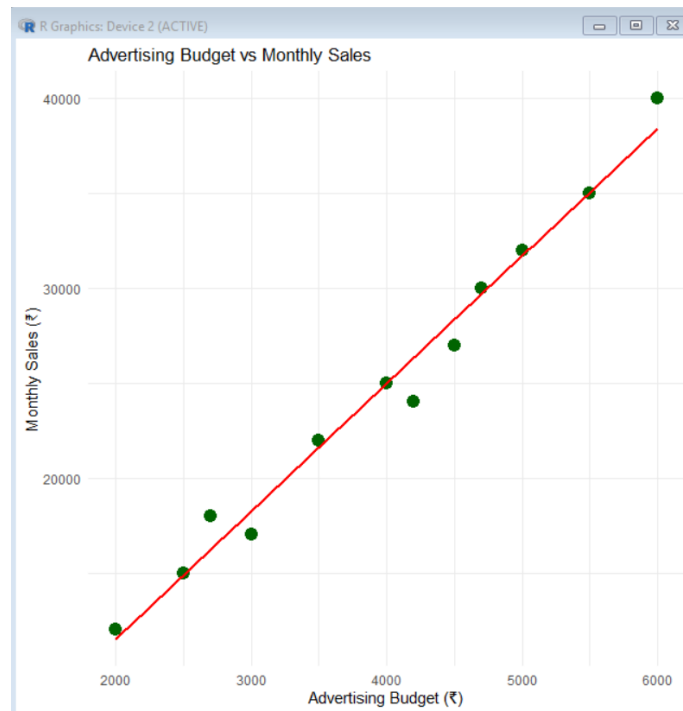
**CODE :**

```
# Load library
library(ggplot2)

# Dataset
advertising <- data.frame(
  Month=c("Jan","Feb","Mar","Apr","May","Jun",
          "Jul","Aug","Sep","Oct","Nov","Dec"),
  Budget=c(2000,2500,2700,3000,3500,4000,
           4200,4500,4700,5000,5500,6000),
  Sales=c(12000,15000,18000,17000,22000,25000,
          24000,27000,30000,32000,35000,40000)
)

# Scatter Plot
ggplot(advertising, aes(x=Budget, y=Sales)) +
  geom_point(color="darkgreen", size=4) +
  geom_smooth(method="lm", se=FALSE, color="red") +
  labs(
    title="Advertising Budget vs Monthly Sales",
    x="Advertising Budget (₹)",
    y="Monthly Sales (₹)"
  ) +
  theme_minimal()
```

**OUTPUT :**



### Insights from Scatter Plot

- There is a **positive relationship** between advertising budget and monthly sales.
- As the advertising budget increases, sales also tend to increase.
- The trend line shows a **strong positive correlation**.
- This suggests that increased advertising may contribute to higher sales.

### Program 4: Interactive Dashboard (Line Chart + Bar Chart)

#### CODE :

```
# Install packages if needed
```

```
# install.packages("flexdashboard")
```

```
# install.packages("plotly")
```

```
library(flexdashboard)
```

```
library(plotly)
```

```
library(ggplot2)
```

```
# Dataset 1
```

```

sales <- data.frame(
  Month=factor(c("Jan","Feb","Mar","Apr","May","Jun",
    "Jul","Aug","Sep","Oct","Nov","Dec"),
    levels=c("Jan","Feb","Mar","Apr","May","Jun",
    "Jul","Aug","Sep","Oct","Nov","Dec")),
  Sales=c(12000,15000,18000,17000,22000,25000,
    24000,27000,30000,32000,35000,40000)
)

```

# Dataset 2

```

products <- data.frame(
  Product=c("Laptop","Mobile","Tablet","Headphones",
    "Smartwatch","Camera"),
  Sales=c(450,700,320,500,280,210)
)

```

# Line Chart

```

line_plot <- ggplot(sales, aes(Month, Sales, group=1)) +
  geom_line(color="blue", linewidth=1.2) +
  geom_point(size=3, color="red") +
  labs(title="Monthly Sales")

```

# Bar Chart

```

bar_plot <- ggplot(products,
  aes(reorder(Product, Sales), Sales, fill=Product)) +
  geom_bar(stat="identity") +
  coord_flip() +
  labs(title="Top Selling Products") +

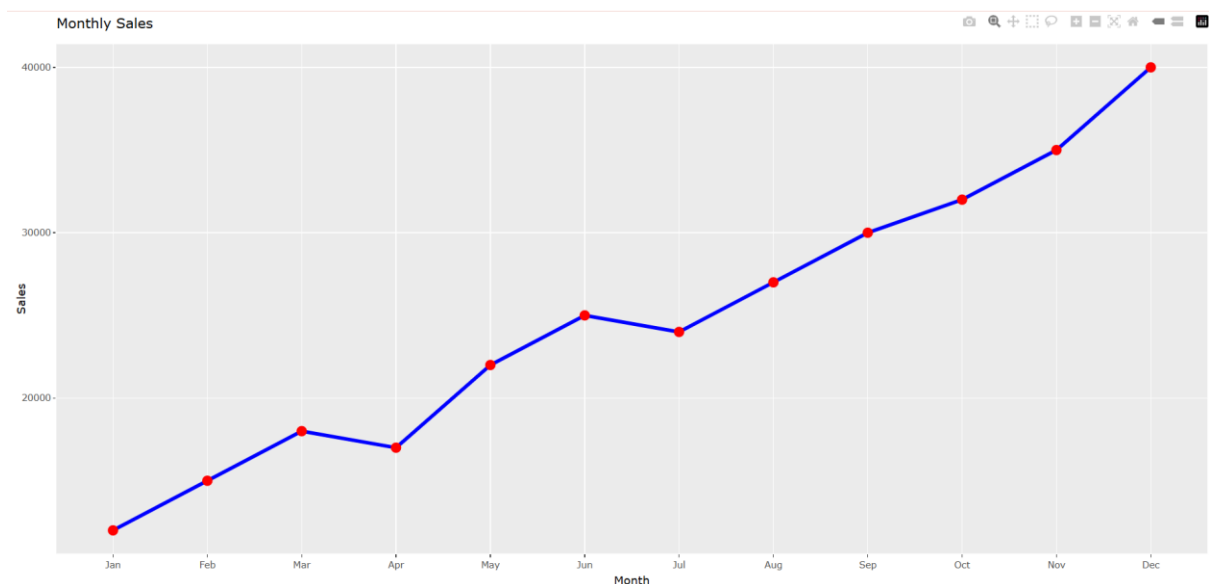
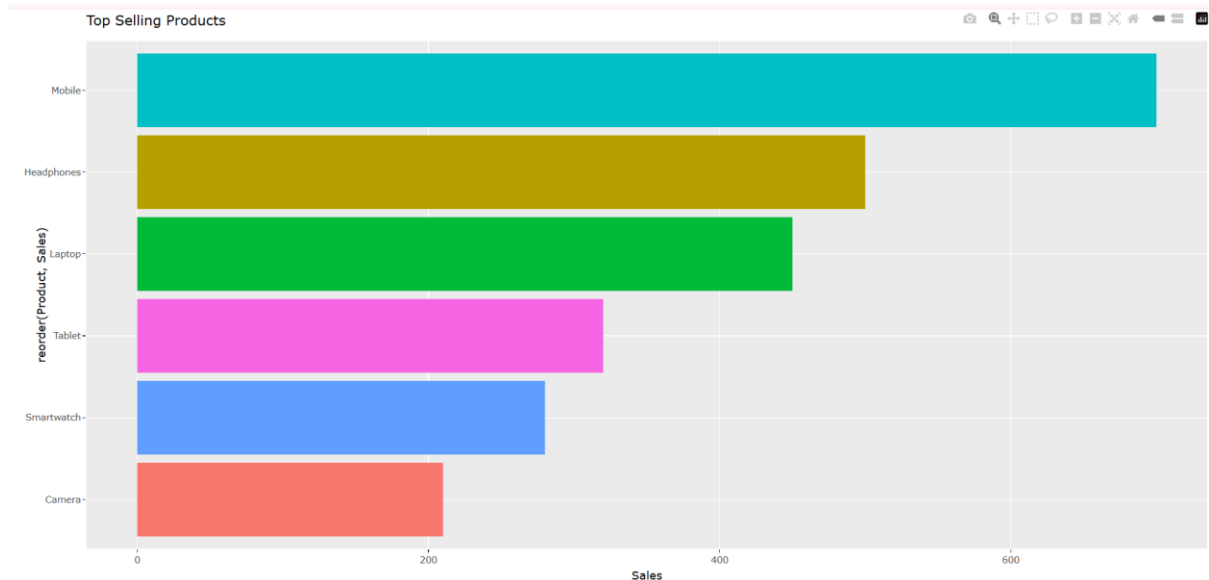
```

```
theme(legend.position="none")
```

```
ggplotly(line_plot)
```

```
ggplotly(bar_plot)
```

OUTPUT :



## SET 2.Program 1: Histogram – Customer Age Distribution

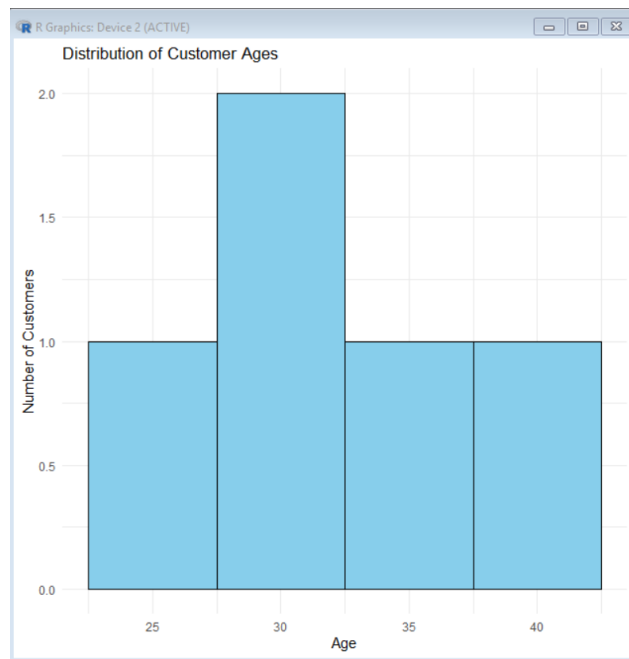
CODE :

```
library(ggplot2)
```

```
customer <- data.frame(
  CustomerID = c(1,2,3,4,5),
  Age = c(25,30,35,28,40),
  Satisfaction = c(4,5,3,4,5)
)

ggplot(customer, aes(x = Age)) +
  geom_histogram(binwidth = 5, fill = "skyblue", color = "black") +
  labs(
    title = "Distribution of Customer Ages",
    x = "Age",
    y = "Number of Customers"
  ) +
  theme_minimal()
```

OUTPUT :



## 2.B : Pie Chart – Customer Satisfaction Distribution

CODE :

```
library(ggplot2)
```



```
# Dataset
```

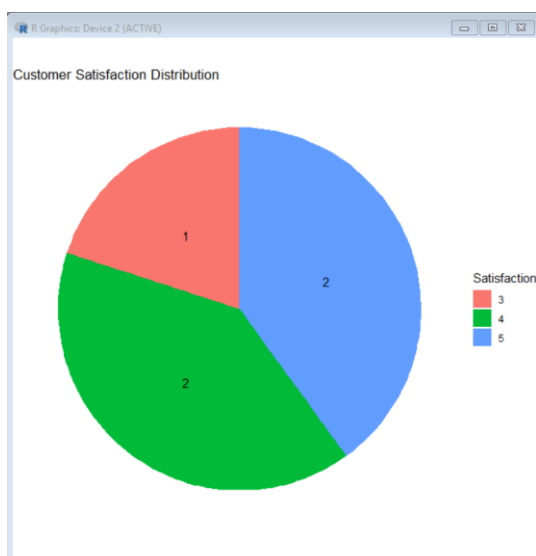
```
customer <- data.frame(  
  CustomerID = c(1,2,3,4,5),  
  Age = c(25,30,35,28,40),  
  Satisfaction = c(4,5,3,4,5)  
)
```

```
# Count satisfaction scores
```

```
pie_data <- as.data.frame(table(customer$Satisfaction))  
colnames(pie_data) <- c("Satisfaction", "Count")
```

```
# Pie Chart
```

```
ggplot(pie_data, aes(x = "", y = Count, fill = Satisfaction)) +  
  geom_bar(stat = "identity", width = 1) +  
  coord_polar("y") +  
  geom_text(aes(label = Count),  
            position = position_stack(vjust = 0.5)) +  
  labs(title = "Customer Satisfaction Distribution") +  
  theme_void()
```



## 2.C: Stacked Bar Chart – Satisfaction by Age Group

### CODE :

```
library(ggplot2)

# Dataset

customer <- data.frame(

  CustomerID = c(1,2,3,4,5),

  Age = c(25,30,35,28,40),

  Satisfaction = c(4,5,3,4,5)

)

# Create Age Groups

customer$AgeGroup <- cut(

  customer$Age,

  breaks = c(20,30,40,50),

  labels = c("21-30", "31-40", "41-50"),

  include.lowest = TRUE

)

# Stacked Bar Chart

ggplot(customer,

  aes(x = AgeGroup,

      fill = factor(Satisfaction))) +

  geom_bar() +

  labs(

    title = "Customer Satisfaction by Age Group",

    x = "Age Group",

    y = "Number of Customers",
```

```
fill = "Satisfaction"
```

```
) +
```

```
theme_minimal()
```



## 2.D .Word Cloud – Customer Feedback

**CODE:** library(wordcloud)

library(tm)

# Customer Feedback

```
feedback <- c(
```

```
"Excellent service and friendly staff",
```

```
"Very satisfied with product quality",
```

```
"Good experience and quick delivery",
```

```
"Friendly support and excellent response",
```

```
"Satisfied with service and quality"
```

```
)
```

```
docs <- Corpus(VectorSource(feedback))
```

```
# Convert to lowercase
docs <- tm_map(docs, content_transformer(tolower))

# Remove punctuation
docs <- tm_map(docs, removePunctuation)

# Remove numbers
docs <- tm_map(docs, removeNumbers)

# Remove common words
docs <- tm_map(docs, removeWords, stopwords("english"))

# Create Term Matrix
dtm <- TermDocumentMatrix(docs)

# Convert to Matrix
m <- as.matrix(dtm)

# Word Frequency
word_freq <- sort(rowSums(m), decreasing = TRUE)

# Word Cloud
wordcloud(
  words = names(word_freq),
  freq = word_freq,
  min.freq = 1,
  random.order = FALSE,
  colors = rainbow(length(word_freq))
)
```

